

1. Administrative & Study Identification

Full Study Title: A Study of Investigational Agents in Participants With Previously Treated Stage IV Nonsquamous Non-small Cell Lung Cancer (NSCLC) (MK-3475-01H/KEYMAKER-U01)
Official Regulatory Title: KEYMAKER-U01 Substudy 01H: A Phase 2, Randomized, Umbrella Study With Rolling Arms of Investigational Agents in Participants With Previously Treated Stage IV Nonsquamous Non-small Cell Lung Cancer (NSCLC) **Short Title/ Acronym:** KEYMAKER-U01 Substudy 01H **Protocol Number:** MK-3475-01H / 3475-01H **Master Protocol:** MK-3475-U01 (KEYMAKER-U01) - NCT04165798 **NCT Number:** NCT06780085 **Posting ID:** 558163847581215514 **Trial Status:** RECRUITING (Currently recruiting participants) **Sponsor/Company:** Merck **Investigational Product:** Ifinatumab deruxtecan (I-DXd), Raludotatug deruxtecan (R-DXd), Docetaxel (Trade names: Docefrez, Taxotere; ATC Code: L01CD02) **Phase of Development:** Phase 2 **Medical Monitor:** Clinical Operations Lead, Merck

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the Objective Response Rate (ORR) per RECIST version 1.1 assessed by blinded independent central review (BICR), as well as safety and tolerability evaluated through the percentage of participants with at least one adverse event (AE) and those who discontinue medication due to an AE. **Secondary Objectives:** To evaluate Duration of Response (DOR), Progression-Free Survival (PFS), and Overall Survival (OS).

2.2 Background & Rationale To address the clinical need for effective treatments in patients with stage IV nonsquamous non-small cell lung cancer (NSCLC) who have experienced disease progression following first-line anti-PD-(L)1 therapy and platinum-based chemotherapy. Antibody-drug conjugates (ADCs) such as ifinatumab deruxtecan and raludotatug deruxtecan are being evaluated against the standard of care (docetaxel) to improve survival and response outcomes in this patient population.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Docetaxel) **Blinding:** Single-blind **Randomization:** Randomized (Umbrella study with rolling arms)

3.2 Planned Enrollment & Duration Target **\$N\$:** 96 participants
Number of Sites: Global (Multicenter including US, Europe, and RoW)
Estimated Study Start: 12-May-2025 **Estimated Primary Completion:** 12-Mar-2032

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically- or cytologically-confirmed diagnosis of Stage IV nonsquamous non-small cell lung cancer (NSCLC). 2. Measurable disease per Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1 as assessed by investigator and verified by BICR. 3. Documented disease progression per RECIST 1.1 after receiving an anti-programmed cell death 1 protein (PD-1)/programmed cell death ligand 1 (PD-L1) treatment and platinum-based chemotherapy. 4. Confirmation per local test report that epidermal growth factor receptor negative (EGFR-), anaplastic lymphoma kinase negative (ALK-), or c-ros oncogene 1 negative (ROS1-) directed therapy is not indicated as primary therapy. 5. Life expectancy of at least 3 months. 6. Eastern Cooperative Oncology Group (ECOG) performance status of 0 to 1 assessed within 7 days before randomization. 7. Provision of an archival tumor tissue sample (not previously irradiated) and tissue from a newly obtained formalin-fixed sample from a new biopsy. 8. Participants with well-controlled HIV (on antiretroviral therapy), HBV (undetectable viral load on antiviral therapy for at least 4 weeks), or HCV (undetectable viral load at screening) are eligible.

4.2 Exclusion Criteria 1. Diagnosis of small cell lung cancer or, for mixed tumors, presence of small cell elements. 2. History of (noninfectious) pneumonitis/interstitial lung disease (ILD) that required steroids, or has current pneumonitis/ILD, and/or suspected ILD/pneumonitis. 3. Active autoimmune disease that has required systemic treatment in the past 2 years. 4. Prior radiation therapy to the lung. 5. Known untreated central nervous system (CNS) metastases, carcinomatous meningitis, or evidence of any leptomeningeal disease. 6. Uncontrolled or significant cardiovascular disorder prior to randomization. 7. Clinically severe pulmonary compromise resulting from intercurrent pulmonary illnesses.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Ifinatamab deruxtecan (I-DXd) or Raludotatug deruxtecan (R-DXd) versus standard Docetaxel. Administered in sequential cycles. **Route of Administration:** Intravenous (IV). **Duration of Treatment:** Until documented disease progression (verified by BICR), unacceptable toxicity, development of a new malignancy, or participant withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care and symptom management medications according to institutional guidelines. **Prohibited:** Receipt of any other investigational agents within 4 weeks prior to intervention, live or live-attenuated vaccines within 30 days before the first dose, and concurrent immunosuppressive therapies for active autoimmune conditions.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent, Medical History, Tumor Imaging (Baseline RECIST), Biomarker Status Confirmation | | **Baseline** | Day 0 | Randomization, First Dose Administration | | **Interim** | Every 3 Weeks (Q3W) | Safety Labs, AE Monitoring, Efficacy Assessment (Tumor Imaging) | | **End of Study** | At Progression/Withdrawal | Final Vital Signs, Exit Interview, IP Accountability, Survival Follow-up |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **Objective Response Rate (ORR):** Percentage of participants with a confirmed complete response (CR) or partial response (PR) per RECIST version 1.1, assessed by Blinded Independent Central Review (BICR). 2. **Safety Profile:** Percentage of participants with at least one Adverse Event (AE). 3. **Discontinuation Metric:** Percentage of participants who discontinued medication due to an AE.

7.2 Secondary Endpoints 1. Duration of Response (DOR). 2. Progression-Free Survival (PFS). 3. Overall Survival (OS).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of AEs and treatment-related AEs (TRAEs) throughout the study via investigator assessments, with a specific focus on incidence of Interstitial Lung Disease (ILD) / pneumonitis. **Serious Adverse Events (SAEs):** Required to be reported to the sponsor and regulatory authorities within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee evaluating patient safety and risk-benefit profile periodically across the umbrella sub-studies.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy analysis; Safety Analysis Set for all safety and

tolerability endpoints. **Sample Size Justification:** Target enrollment of approximately 96 subjects is planned to provide sufficient statistical power ($\alpha = 0.05$, $\text{Power} \geq 80\%$) to compare the ORR of the ADCs versus Docetaxel. **Handling of Missing Data:** Participants withdrawing prior to disease progression will be censored at the date of their last evaluable imaging assessment for PFS and DOR analysis (Kaplan-Meier estimates).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits by clinical research associates to verify protocol adherence, informed consent integrity, and data accuracy. **Audit Trail:** Electronic Case Report Form (eCRF) utilizing strict audit trails, timely data clarification forms, and scheduled database locks.

11. Study Identifiers & Governance

Primary ID: MK-3475-01H **Master Protocol Registry:** NCT04165798
Registry Number: NCT06780085 **Posting ID:** 558163847581215514
Sponsor/Lead Organization: Merck **Study Title:** KEYMAKER-U01
Substudy 01H Status: RECRUITING

12. Clinical Framework (Oncology Specific)

Disease Areas: Carcinoma, Non-Small-Cell Lung **Preferred UMLS Name:** Non-squamous cell lung carcinoma **Diagnosis Threshold:** Histologically or cytologically confirmed metastatic NSCLC with disease progression following anti-PD-(L)1 and platinum-based chemotherapy **Medical Methodology:** Antibody-Drug Conjugates (ADC) vs. standard-of-care chemotherapy **Optimization Target:** Improvement in Objective Response Rate (ORR) and Progression-Free Survival (PFS)

13. Study Procedures & Methodology

Management Pathway: Sequential IV infusion of assigned therapeutic arm every 3 weeks (Q3W) until disease progression. **Education Protocol (HCP):** Site initiation visits and ongoing investigator meetings regarding ADC toxicity management, particularly ILD monitoring. **Education Protocol (Patient):** Strict informed consent process with ongoing education regarding reporting of pulmonary symptoms. **Quality Audit Mechanism:** Centralized imaging review (BICR) for all tumor assessments to prevent local investigator bias.

14. Observation & Follow-Up Schedule

Total Duration: Until disease progression followed by survival follow-up (up to estimated completion in 2032) **Onsite Visit Cadence:** Every 3 weeks (Q3W) coincident with treatment dosing **Remote Monitoring Frequency:** Between-cycle remote contacts as clinically indicated **Checklist Review Interval:** Regular tumor imaging assessments per RECIST 1.1 guidelines every 9 weeks (standard oncology cadence)

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				